

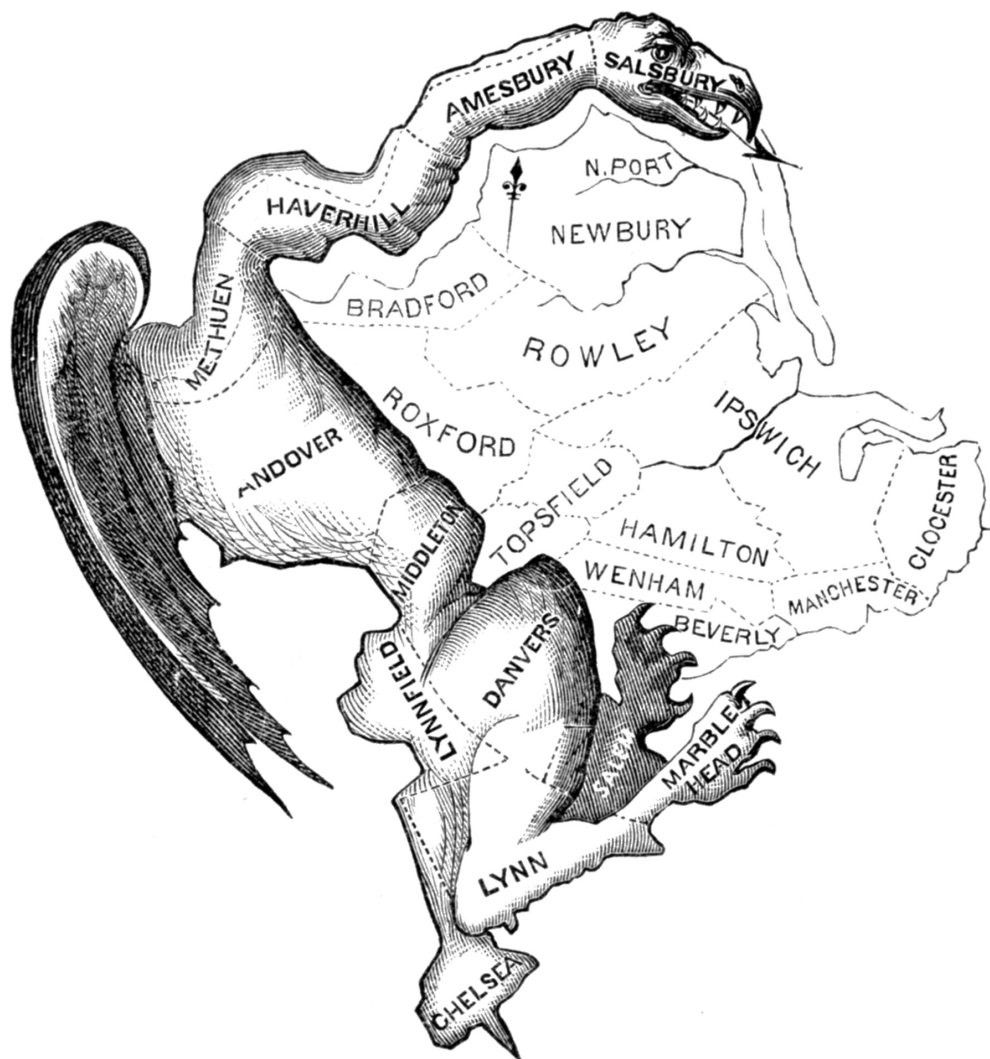


Figure 1: Gerry-Mander, Harbinger of Inequality

Elections in Reverse

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May 4, 2026

1 Introduction

Gerrymandering in the United States usually involves “packing” opposition voters into few districts or “cracking” them across many to dilute their influence. It is a strategic act to maximize the power of parties or individuals by minimizing the voting power of their opposition. In 2013, the Supreme Court removed some restrictions on this tactic with the landmark *Shelby County v. Holder* decision. The decision renders Section 5 of the Voting Rights Act (VRA) unenforceable, effectively eliminating it (Brennan Center for Justice 2023). Section 5 established the preclearance requirement, a safeguard which required specific jurisdictions with histories of racial discrimination to obtain federal preclearance before changing voting laws; the goal was to end discriminatory voting changes before they went into effect rather than relying on lawsuits after the fact. Without preclearance, states and counties with a history of racial discrimination can implement voting laws and maps with little oversight, enabling these states to draw maps which limit minority voices and discriminate against marginalized communities where they could not before. This paper aims to examine the impacts of the *Shelby* decision on minority representation in the U.S. as it pertains to electoral fairness. We argue that the decision negatively impacted minority voting power by demonstrably increasing cracking/packing across the *Shelby* decision, and that states are incentivized to gerrymander with respect to race even when they do not directly gerrymander with respect to party.

The effects of the *Shelby* decision are part of a larger trend in U.S. politics (Bernini et al. 2024). On April 29th 2026 the *Shelby* decision was reinforced by the Supreme Court in *Louisiana v. Callais*, which decided that the protections of the VRA no longer apply due to white and Black parity in “two of the five most recent Presedential elections.” This is a direct continuation of the logic in *Shelby*, characterized by Justice Ruth Bader Ginsburg’s dissent from 2013: “Throwing out preclearance when it has worked and is continuing to work to stop discriminatory changes is like throwing away your umbrella in a rainstorm because you are not getting wet.” The decision specifically affects Louisiana’s revised district map, which as a result of *Callais v. Landry* (2024) included two majority-Black districts out of Louisiana’s six. *Callais v. Landry* ruled the original 2020 congressional map an unconstitutional racial gerrymander under the VRA, finding that race was improperly prioritized in the creation of the plan. With the overturn of *Callais v. Landry* in 2026, racial gerrymandering has been effectively ruled constitutional.

This trend against the VRA is harmful in two principal ways. Firstly, removing safeguards to protect minority voting power will inevitably have negative impacts on minority voting power. Ending preclearance over covered states gives leeway to individuals and institutions with the capacity to discriminate, and enables them to strategically restrict the voting power of minorities towards partisan gain or racial prejudice. Secondly, no federal law prohibits states from undertaking partisan gerrymandering; it is entirely up to states and Congress to regulate redistricting. One of the only impacts on partisan gerrymandering has been the VRA, which can indirectly prevent some partisan gerrymandering due to the fact that racial and ethnic minorities largely identify with the Democratic party (Protect Democracy 2023). Race is the best predictor of partisanship over any other demographic factor; efforts to gerrymander on the basis of partisanship often overlap with demographic gerrymandering,

and imply that the *Shelby* decision would have significant impacts not only on minority representation, but also partisan gerrymandering.

2 Background

The *Shelby* decision has dangerous implications for minority representation in the U.S. The literature on the *Shelby* decision is rich, and usually fall into two major analytical categories. First is the analysis of voting laws, specifically changes to voting procedure post-*Shelby*. Since 2013, covered states have surfaced new laws restricting voting, including voting ID laws, limits on early voting, and same-day registration. Many of these changes have been proven to be discriminatory in intent; North Carolina notoriously instituted new requirements which were struck down in 2016 with *The North Carolina State Conference of the NAACP v. McCrory*, having proved that the new requirements “target[ed] African Americans with almost surgical precision.” Despite the wave of controversial new voting laws post-*Shelby*, studies on the effects of new voting laws have had mixed results. Some have documented a negative effect on voter turnout (Hood and Buchanan 2020). Conversely, Cantoni and Pons 2021 find that ID laws had a very limited effect on Black registration and turnout, finding that mobilization campaigns targeting minorities were successful in offsetting efforts to disenfranchise them. This result is crucial in our analysis. From a strategic perspective, it makes sense under the issue attention cycle (Downs 1972) that these campaigns might lose inertia over time as the laws remain entrenched in public policy. Even if a change in partisan representation is not immediately observed, these changes have long-reaching effects that both disproportionately hurt minorities and prop up the voices which would suffer from their representation.

The second category of analysis focuses on redistricting fairness and representation, which we will focus on. A vital detail in any analysis of gerrymandering is some definition of fairness. Fairness to the people is the cornerstone of the U.S. Constitution, and voting is arguably the most important mechanism behind that fairness. The “fairest” possible system might be one in which the vote of any person is worth just as much as any other person. Here is where the trouble begins: this restricts the power of a “state” as an entity that should have independent and comparable power to other states. Under this definition, a state with more people is arguably more influential in federal decision-making than a state with fewer people. Famously, this dilemma is what the U.S. Congress and Senate aim to alleviate, but it also creates complexity around the seemingly simple idea of one person, one vote.

Gerrymandering, while termed in 1812, is an unfortunate tradition nearly as old as the United States itself and has been studied for at least as long. Steering state districts in favorable directions for a political party is a natural tendency for political institutions wishing to maintain power and influence. A benevolent institution which aims to protect single votes still operates in a representative government, requiring a degree of information loss from single vote counts. The tools for district control in the U.S. have always rested in the laps of incumbent interests. Gerrymandering may be profoundly undemocratic, but it is also unsurprisingly widespread in its usage across party and ideology. In the context of millions of people, fairness, especially in redistricting, has historically proven difficult due to these incentives. Different groups and political parties aim to limit the power of their

opposition, taking whatever definitions of fairness might be most convenient to maintaining or implementing a power dynamic (Becker and Gold 2022).

Another straightforward idea of fairness is political representation. Presently U.S. politics is dominated by two political parties, so with relatively little information loss, ensuring partisan symmetry across population and representation is an ideal metric (Gelman 2002). In this formulation, fairness of a particular redistricting plan can be quantified by the relationship between a proportion of votes a party receives and the proportion of seats it wins. If a state has ten seats and a perfect partisan split, one might expect that both parties have five seats each. There are two versions of this formulation, one comparing the seat share each party would win if support were shifted to 50% per party, and the second comparing with a specific symmetry parameter $\bar{p}$. This enables us to evaluate and compare plans with a mathematically rigorous definition of electoral responsiveness and bias. In addition to partisan bias, political scientists have developed a quantifier termed the "efficiency gap" (Warrington 2018). Partisan bias quantifies the degree to which parties would win different fractions of the total seats with the same original votes. The efficiency gap relies on the concept of a *wasted vote*, these being votes that are cast over the majority required to win an election. We can see this in a simple toy example detailed in Figure 2. Two districts 1 and 2 have ten voters each. One district is won by party A with 7 votes and the other by party B with 6 votes. In district 1, every vote over 50% of the population is 'wasted.' An optimal outcome for party A would be to redistrict 3 of its voters into District 2, where, despite holding 50% of the partisan vote across districts, party A would hold 100% of seats.

Another requirement for our analysis is that we have a frame of reference to compare different redistricting plans. While we can quantify fairness with regard to the voting populous or partisan representation atomically using partisan bias or the efficiency gap, these plans are often drawn to maximize a power dynamic with respect to a set of rules which vary by state. There is no baseline for comparison. The ALARM Project (McCartan, Kenny, Simko, Garcia, et al. 2022) proposes

District	Party A	Party B
District 1	7	3
District 2	4	6

Figure 2: Two-district vote distribution (10 voters each)

a simulation-based approach to contextualize congressional district plans within the rules of a state, enabling state-specific district simulations. The authors simulate possible district plans based on the rules of a state, considering geography, demographics, institutions and redistricting requirements that constrain the space of possible maps. Drawing possible simulated plans thousands of times, they are able to evaluate different quantifiers of fairness over the generated set. This formulation allows for a counterfactual "ground truth" distribution of possible plans which can be used to contextualize existing plans, and has been used extensively in further literature.

The simulated plans take into account existing redistricting rules per-state, but lack more complicated rules from the Voting Rights Act (VRA). These ensembles are still effective baselines, but are ineffective for contextualizing proposed plans with regard to prohibitions against racial gerrymandering and minority vote dilution, specifically the "Gingles factors"

which mandate the creation of minority-opportunity districts. This is ameliorated with “VRA-conscious” ensembles that are designed with these constraints (Becker, Duchin, et al. 2021), advancing the ALARM Project’s framework by integrating the legal requirements of the VRA into simulation algorithms.

The ALARM Project offers a comprehensive simulation-based framework to contextualize redistricting plans, and offers excellent analysis of district fairness in the context of neutral counterfactual alternative districts. While district demographics are a facet of their Sequential Monte Carlo (SMC) algorithm (and therefore existing plans that deviate from the neutral counterfactuals can be said to possibly deviate from VRA requirements), they do not offer specific analysis of the effects of current district plans on minority group representation. There exist specific case studies Gay 2007 and general improvements on simulating districts Becker, Duchin, et al. 2021, but there is limited material assessing the statistical impact on marginalized groups from these district plans on a federal level, as well as targeting covered states from the VRA post-Shelby decision.

McCartan et al. (McCartan, Kenny, Simko, Ebowe, et al. 2025) utilize the ALARM framework to evaluate the pre and post-Shelby cycles. They perform a DiD experiment with a treatment consisting of states that implemented new redistricting reforms between cycles, with the control consisting of states that maintained their existing map-drawing processes. They compare these to the ALARM Project’s neutral baseline. Our proposed framework performs a DiD with the control as the non-covered states (which presumably would see little change) and the treatment group as the covered states, which presumably would change more in response to the Shelby decision. While these authors evaluate redistricting outcomes as a function of institutional reform type, they do not study the effect of the removal of federal preclearance itself on minority representation.

Komisarchik and White (Komisarchik and White 2020) provide an empirical audit of the *Shelby* decision’s effects on minority voter participation. Utilizing a DiD design comparing previously covered jurisdictions with similar non-covered areas, they study a possibly widening gap between Black and White voter turnout following the ruling, but find limited effects. Crucially, this work’s thesis demonstrates that the effects of *Shelby* extend beyond the physical lines of redistricting maps; the removal of federal preclearance enabled a suite of electoral changes that target minority groups and can systematically depress minority participation, establishing that structural legal protections are directly linked to the exercise of electoral power. These authors establish that *Shelby* depressed minority voter participation through procedural changes, but do not examine whether the the degree of racial packing in enacted maps shifted in response to the removal of federal oversight.

A critical institutional backdrop to post-*Shelby* redistricting is the Department of Justice’s “Max-Black” strategy, pursued aggressively during the 1990s redistricting cycle under VRA Section 5 preclearance authority. Under this interpretation of the non-retrogression principle, the DOJ pressured covered jurisdictions to maximize the number of majority-minority districts, often producing “super-majority” districts in which Black populations were concentrated well beyond what was necessary to elect a minority-preferred candidate (Lublin 1997). Scholars have further noted that some covered state legislatures exploited VRA compliance as a partisan pretext, using the mandate to pack reliable Democratic voters into

fewer districts while engineering safer Republican seats in the surrounding areas.

Majority-minority districts are further explored by [Trautman and Smith 2025](#), analyzing how the 2018 and 2022 midterm redistricting cycles influenced the turnout and mobilization of Black voters. The study finds that the structural configuration of a district, specifically the degree to which minority voters are packed or cracked, impacts grassroots mobilization and engagement with the electoral process. This suggests that the harm of unfair redistricting is not limited to a loss of legislative representation, but extends to a structural dampening of democratic participation itself. This is further studied in majority-minority districts by [Gay 2007](#), who examines the trade-offs of packing minority voters into dedicated districts with respect to the responsiveness of legislators in neighboring ones. She argues that while such districts ensure descriptive representation, they simultaneously “liberate” surrounding white legislators from having to appeal to minority interests in order to win. This dynamic, which she terms legislating without constraints, creates a scenario in which the substantive policy influence of a minority group may decrease even as their nominal seat count remains stable.

The downstream consequences of this packing strategy on minority groups is dangerously substantial. [Lublin 1997](#) argues that while majority-minority districts reliably increase the number of Black officeholders, they produce a perverse effect on the surrounding area: by drawing Black voters who lean heavily Democratic out of adjacent districts, those districts become more likely to elect Republicans whose policy agendas may be misaligned with minority interests. Because districts must be equally populous under federal law, the creation of high-concentration majority-minority districts in covered states operates as a zero-sum reallocation of minority influence across the broader political map.

A vital consideration in our analysis should also be how these maps are designed. In practice, district maps are drawn by whichever partisan body controls the state legislature, typically with the assistance of professional mapmakers and Geographical Information System software. One such mapmaker was Thomas Hofeller, a renowned gerrymanderer for the Republican party. While his work in states like North Carolina is more famous, his gerrymandering strategy was adopted professionally by the Republican party across the south through his company, “Geographic Strategies.” From some points of view, Hoefeller could be described a proponent for the VRA: he was a loud advocate for Black majority districts across the south. This strategy, while initially helping to elect more Black representatives, also served to pack Democratic-leaning Black voters into fewer districts and ultimately constrict their voting power by making the surrounding white districts more reliably Republican ([Bouton et al. 2023](#)). After his death, a trove of his personal files were released which detailed some of the data and analysis used to develop these maps, exposing the extensive data on race and other demographics which work behind the scenes to restrict minority voting power and give Republicans a partisan edge ([Common Cause v. Lewis 2019](#)). The Hoefeller example demonstrates the ways racial and partisan gerrymandering are often linked, and that even when not specifically targeting partisan outcomes, the process of drawing electoral maps often specifically target marginalized groups using very specific data analytics.

3 Case Study: Louisiana

A motivating example for this study is the state of Louisiana as the focus of *Louisiana v. Callais* (2026). The decennial 2020 district plan exemplifies what we aim to study, that minority packing may be more likely in covered states, and that these states might increase across the *Shelby* decision. The state also serves as an example of the mechanics of racial and partisan packing, and will motivate our chosen features of importance in our analysis.

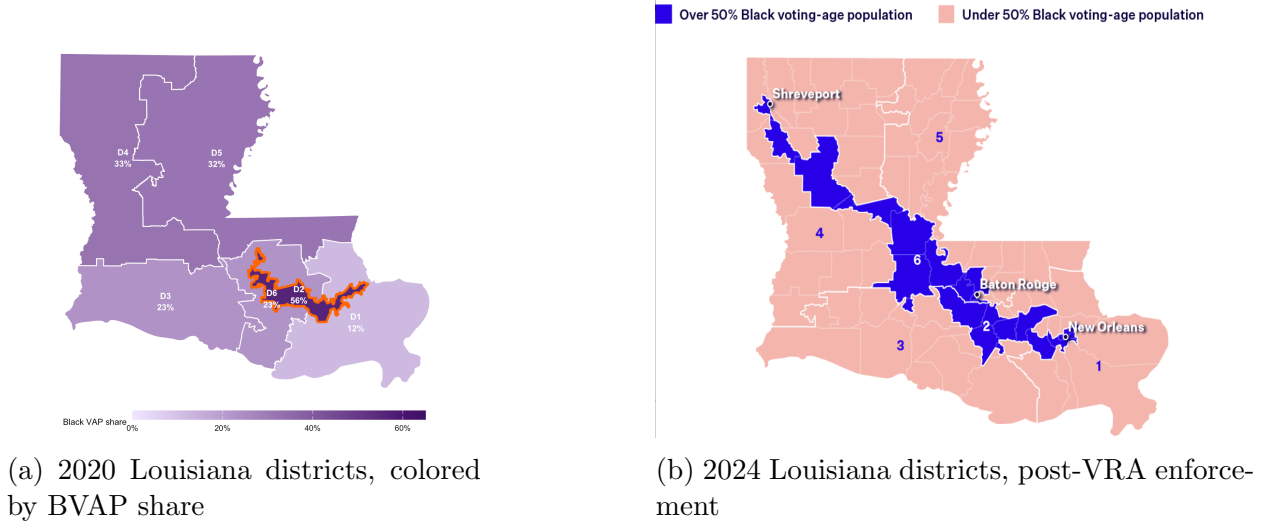


Figure 3: Louisiana district maps pre-*Callais* (2026)

Following the 2020 decennial census, the Louisiana State Legislature enacted a map (Figure 3a) that maintained only a single majority-Black district (the 2nd District), despite Black residents comprising approximately one-third of the state’s population and registered Democrats making up 36.50% of the electorate. This 2020 plan faced intense legal scrutiny in *Ardoin v. Robinson* (2023) and *Callais v. Landry* (2024), characterized by the extreme concentration of minority voters into a single district to prevent the formation of a second competitive seat. With *Callais v. Landry*, the plan was redrawn in order to comply with Section 2 of the VRA. This revised plan, seen in Figure 3b, introduced a second majority-Black district (the 6th District), stretching from Shreveport to Baton Rouge and significantly altering the state’s *max BVAP* and *efficiency gap* metrics. With *Louisiana v. Callais* (2026), the state is now free to return to the original plan.

As a covered state, Louisiana illustrates the dangers of removing federal oversight from historically discriminatory institutions. The state has a documented and long history of racially motivated redistricting, with enacted maps that have been successfully challenged for packing and cracking Black voters in violation of the VRA. Its redistricting record also elucidates an important question for our analysis: what features of a district map, constitute a violation of the VRA or a meaningful departure from fairness?

4 Analysis

We focus the scope in our analysis to estimations of district packing, which we extract from simulated and real districts by taking the maximum over all districts. This gives a simple, effective metric across our statistics. We specifically focus on two primary metrics computed for each enacted plan. The first is *max BVAP*, the maximum Black voting-age population (BVAP) share across all districts in a plan, which serves as our measure of racial packing: the degree to which Black voters have been concentrated into a small number of districts. The second is *enacted EGAP*, the efficiency gap, which indexes partisan packing and cracking jointly. By focusing on the *maximum BVAP* and *efficiency gap*, we isolate the ‘packing’ mechanism of gerrymandering, where minority or partisan groups are concentrated to a degree that exceeds what would be expected under neutral geographic and demographic constraints with respect to the Black population and partisan packing. We do not separately assess cracking in this analysis as these statistics are not as naive to track as simple maximums, though we acknowledge that this would be an interesting area for future research; this is particularly true considering how Louisiana’s 2020 plan cracks the BVAP (*Callais v. Landry*), which is harder to visually identify or computationally analyze when comparing the revised 2024 plan to the original.

The demographic and electoral data underlying our analysis come from the ALARM Project’s redistricting data files (McCartan, Kenny, Simko, Garcia, et al. 2022), which merge and tidy two primary sources. Racial and ethnic breakdowns of both the total population and the voting-age population are drawn from the 2020 decennial Census and 2010 decennial Census, tabulated at the precinct level. These are joined to precinct-level election returns from the Voting and Election Science Team (VEST), which provide average vote counts for Democratic and Republican candidates across statewide races. Where 2020 precinct boundaries are unavailable, census block-level data are provided instead.

Our analysis is two-pronged. We first evaluate the differences between covered and non-covered states within the 2020 district plans build off of our Louisiana case study and provide a basis for our argument: covered states are more likely to pack one max district than non-covered states, in both minority and partisan packing. This analysis mainly focuses on the 2020 district data and contextualizes it with 2010 figures, which we focus on to highlight existing disparities as context for our test. Second, we detail several statistical experiments to evaluate the effect of the *Shelby* decision. We evaluate the difference between covered and non-covered states across the *Shelby* decision to determine whether or not minority packing increased for covered states, and theorize what effects it may have. The states we will examine include:

- Fully covered (n=8): AL, AZ, GA, LA, MS, SC, TX, VA
- Partially covered (n=5): CA, FL, NH, NY, NC
- Not covered (31 states with ALARM data)

Because redistricting outcomes are heavily contingent on a state’s unique geography and demographic distribution, raw metrics like the *Efficiency Gap* or *BVAP* percentage cannot be directly pooled in a regression. To ensure comparability across states, we assess departures

from the simulated distribution using two complementary simulation-relative statistics: a z-score and a percentile rank. Within simulation metrics, we define the former as the enacted plan’s deviation from the simulation mean in units of standard deviation. In demographic metrics on a per-state basis, we define it as the enacted plan’s BVAP with regard to the population of the state atomically. We define the latter within simulation metrics alone, as the fraction of simulated plans falling below the enacted value. These metrics enables comparisons across states with different populations, standardizing the magnitude of “extremeness” across jurisdictions. This approach transforms state-specific results into a common metric of “departure from neutrality,” allowing for a statistically sound comparison between states with vastly different baseline populations and partisan leans.

4.1 2020 Plan Baselines

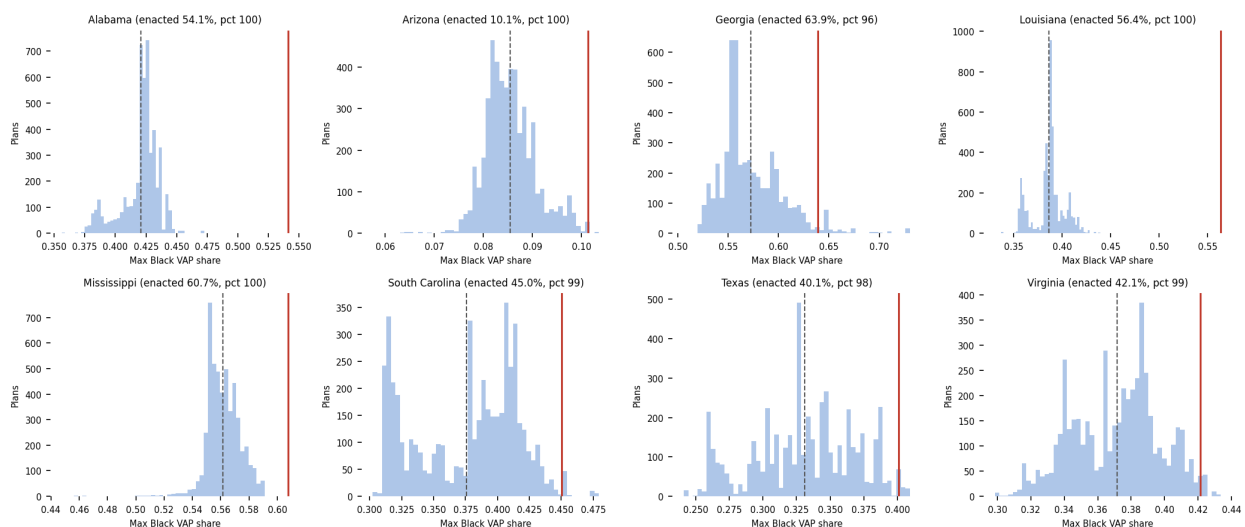


Figure 4: Histogram of covered states’ simulated max BVAP with enacted plan as a red vertical line.

We first utilize the ALARM Project’s simulated district data and 2020 precinct-level demographic data to construct district-level minority voting power metrics, utilizing a Gingles packing excess measure defined as the surplus Black Voting Age Population (BVAP) above the 55% safe-harbor threshold that is typically sufficient to elect a minority-preferred candidate. We use this to compare covered and non-covered states and establish a basis for our claim: covered states are more likely to pack than non-covered states in single years. In Figure 4, we can see that the *max BVAP* share across all districts in covered states is far from the neutral simulation distribution, greater than 95th percentile across all states. This means that the observed levels of racial concentration in these jurisdictions are not the product of non-partisan geographic constraints. Instead, the enacted maps are statistical outliers that concentrate Black voters into “super-majority” districts at rates significantly higher than what would occur under a race blind, neutral redistricting process.

This trend can also be compared across states. Figure 5 details a national choropleth of racial packing in 2020 congressional redistricting plans, measured against neutral simulation

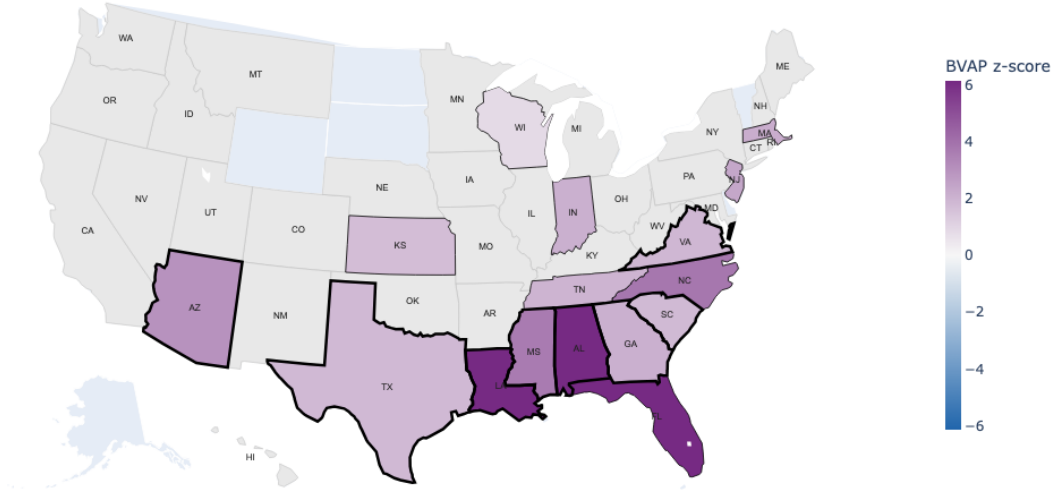


Figure 5: 2020 z -scored max BVAP packing across the US

baselines from the ALARM redistricting project. For each state, approximately 5,000 theoretically race-blind redistricting plans were simulated under their neutral algorithmic process. Per state, we record the max BVAP share across the congressional districts of each enacted plan, and compare it to the distribution of that same statistic across the simulated plans. States are colored only where the enacted plan constitutes a statistically significant outlier (with regard to the state’s simulated plans alone), which we define to be if fewer than 5% of simulated plans produced a maximum Black district BVAP share equal to or greater than the enacted plan’s ($p \leq 0.05$). Color intensity reflects the BVAP z -score, with purple indicating greater packing than neutral plans would predict and blue indicating less. States shown in gray are not statistically distinguishable from neutral redistricting with regard to BVAP. States with thick black borders were fully covered. This plot is central to our argument—when corrected for population, covered states pack Black voters into super-majority districts where non-covered states deviate less.

Figure 6 tracks the trajectory per-state of the max BVAP congressional district between the 2010 and 2020 redistricting cycles. For each state, we identify the single enacted district with the highest Black voting-age population (BVAP) share - the “rank-1” district - and plot an arrow from its position in 2010 to its position in 2020. The gray cloud shows a sample of districts drawn from neutral, race-blind ALARM simulations for the 2020 cycle, providing a baseline for what the rank-1 district might look like under non-partisan redistricting. The left panel shows the eight fully-covered states under the VRA prior to the *Shelby* decision; the right panel shows all other states with data available in both cycles. Arrows moving rightward indicate increased racial concentration in the most-Black district across cycles; arrows moving downward indicate reduced Democratic presidential performance in that district, consistent with a dilution strategy. While the max BVAP district across 2020 and 2010 may not be the same, it is nonetheless interesting to observe that over the *Shelby* decision max BVAP districts covered states do not increase BVAP share, and that many decrease their share.

4.2 Differences-in-Differences Across *Shelby*

Our central inquiry examines whether the trajectory of district packing from 2010 to 2020 differed significantly between fully VRA-covered states and non-covered states. Specifically, we ask whether congressional districts in Section 5 covered states are more racially packed than would naturally arise from race-neutral redistricting, and whether the *Shelby* decision exacerbated this difference between covered/non-covered states. We evaluate this for two targets: demographic packing (defined by the creation of majority-minority districts that hyper-concentrate minority voting power) and political packing (measured via the efficiency gap). Here we again utilize the ALARM Project’s 50-state simulations.

To evaluate these shifts, we define our temporal and legal variables to precisely isolate the impact of the *Shelby* decision. We define our *Post*-treatment period as the 2020 redistricting cycle. By comparing the 2010 and 2020 decennial cycles, we capture the atomic redistricting events immediately preceding and following the ruling, minimizing noise from off-cycle or court-mandated mid-decade redraws and ensuring maps are compared under the same fundamental constitutional requirements. Regarding legal status, our *Treated* variable focuses on the VRA Section 5 coverage formula active prior to *Shelby*. We opt to categorize partially-covered states (e.g., CA, FL, NY) as non-covered due to their ambiguous signal at the congressional level. Sensitivity analyses confirm our results are robust to this specification, as well as to the inclusion of state fixed effects and raw efficiency gap outcomes; we find no significant difference in coefficients whether partially covered states are included in the control group or excluded entirely.

We use a Difference-in-Differences (DiD) experiment to estimate packing changes in two groups across the pre and post-*Shelby* era, where *Post* is 1 for the 2020 group and 0 for the

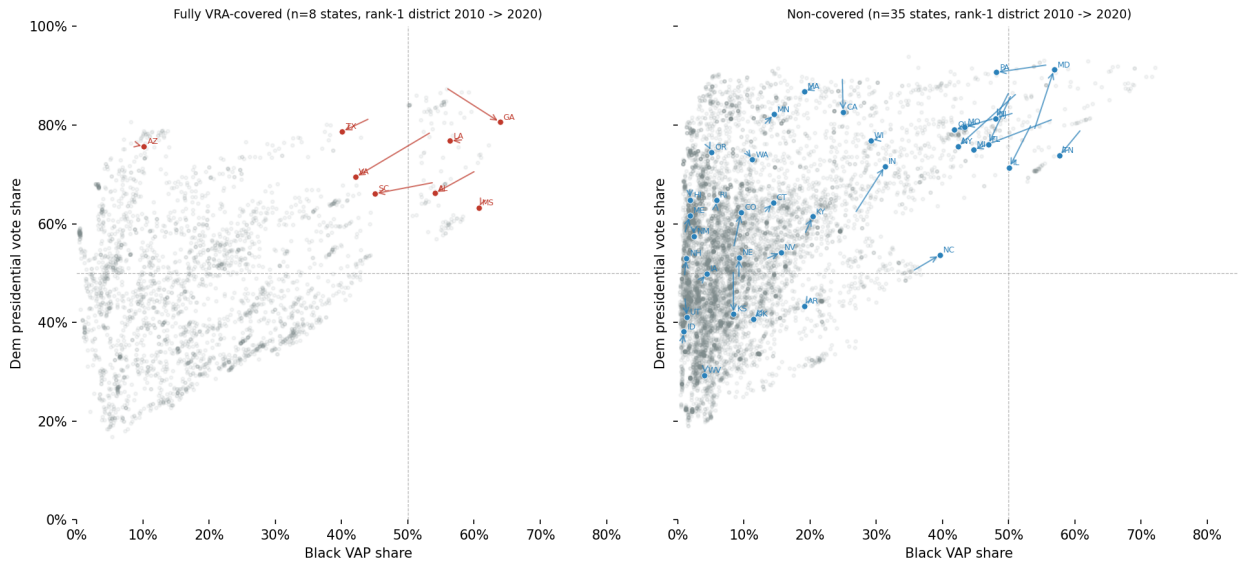


Figure 6: *Max BVAP* districts plotted by dem presidential vote share and BVAP share. **Left** covered, **right** not covered. Colored dots are the maximum BVAP districts per state, annotated with their change from 2010 in the same space.

2010 group, $Treated$ is 1 for fully-covered VRA states (“AL”, “AZ”, “GA”, “LA”, “MS”, “SC”, “TX”, “VA”), and β_3 is the DiD estimator:

$$\Delta packing = \beta_0 + \beta_1 \times Treated + \beta_3 \times (Post \times Treated) + \epsilon \quad (1)$$

We also perform the same experimental setup to predict the efficiency gap, utilizing the following formulation:

$$\Delta egap = \beta_0 + \beta_1 \times Treated + \beta_3 \times (Post \times Treated) + \epsilon \quad (2)$$

From [Table 1](#), the efficiency gap results are statistically insignificant. The primary finding from this Difference-in-Differences model is a null result regarding partisan packing/cracking. The data does not show a statistically significant difference in how the Efficiency Gap (z -score) changed between VRA-covered ($Treated$) and non-covered states following the 2020 redistricting cycle ($Post$ -*Shelby*). The DiD estimator ($\beta_3 = 0.904, p = 0.408$) fails to reach significance, suggesting that any shift in partisan symmetry within VRA-covered states was not meaningfully different from the shifts observed in the rest of the country. This null finding is evidence against our hypothesis, that the *Shelby* decision increased packing. While the legal landscape for minority representation shifted dramatically, the “partisan efficiency” remains un-impacted across the decennial plans.

Variable	Estimate	Std. Error	t	p
Intercept	0.434	0.357	1.215	0.229
Post	-0.818	0.505	-1.621	0.110
Treated	0.840	0.768	1.094	0.277
Post $\times$ Treated	0.904	1.086	0.833	0.408
Observations	74			
R^2	0.106			
Adjusted R^2	0.068			
Residual Std. Error	1.922 (df = 70)			
F -statistic	2.767* (df = 3, 70)			

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Dependent variable: simulation z -scores for the efficiency gap.

Table 1: Difference-in-Differences: Efficiency Gap (z -score)

The Difference-in-Differences analysis of *max BVAP* in [Table 2](#) demonstrates that the primary distinction in racial packing is not the change over time, but the persistent and extreme baseline of the treated states. While the *Shelby* decision removed the federal preclearance requirement, the “Post $\times$ Treated” coefficient suggests that this legal shift did not lead to a statistically significant surge in further packing beyond the already extreme levels established in the 2010 cycle. Instead, the data suggests that covered states have maintained a “packing equilibrium.” The z -score of 6.010 for the Treated variable confirms that these jurisdictions were already outliers in the 2010 cycle, likely a result of the “no-retrogression” standard

of Section 5 which, ironically, often incentivized the maintenance of super-majority districts pushed by gerrymandering entities like Thomas Hoefeller. The lack of a significant DiD effect ($p = 0.154$) indicates that the post-2020 maps in these states continue to be extreme outliers relative to neutral simulations, regardless of the change in their legal oversight status.

Variable	Estimate	Std. Error	t	p	
Intercept	0.229	0.453	0.507	0.614	
Post	-0.069	0.635	-0.109	0.914	
Treated	6.010	0.987	6.091	<0.001	* * *
Post $\times$ Treated	-2.005	1.393	-1.439	0.154	
Observations			77		
R^2			0.428		
Adjusted R^2			0.404		
Residual Std. Error			2.480 (df = 73)		
F -statistic			18.17*** (df = 3, 73)		

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Dependent variable: simulation z -scores for maximum BVAP share.

Table 2: Difference-in-Differences: Max BVAP (z -score)

4.3 Ideology and Packing

Building off of our analyses, we examine the effect of packing these districts on ideology of the candidates which represent them. To accomplish this, we utilize the DW-NOMINATE (Dynamic Weighted Nominal Three-step Estimation) metric, which acts as a political spatial mapping, placing every member of congress on a spectrum based on their roll-call voting records (Poole and Rosenthal 2007). The motivation behind this experiment is that majority-minority districts (often the result of packing) are less competitive, which lead to idiosyncratic representatives Gay 2007. Beyond restricting the power of minorities, we also want to understand how these representatives behave ideologically, and whether or not racial packing impacts ideological scores.

By placing every member of the 117th Congress on a spatial map ranging from -1 (most liberal) to 1 (most conservative), we can determine if the packing identified in our simulation-based z -scores translates into a statistically significant shift in ideological extremity. This allows us to assess whether the *Shelby* decision primarily affected the number of minority-influenced seats or if it also fundamentally altered the ideological extremity of representation within those districts. We use the following setup:

$$\text{NOMINATE} = \beta_0 + \beta_1 \times \text{BVAP} + \beta_2 \times \text{Treated} + \beta_3 \times (\text{BVAP} \times \text{Treated}) + \beta_4 \times \text{Party} + \epsilon \quad (3)$$

Our analysis of DW-NOMINATE scores reveals that the impact of the *Shelby* decision has limited impact on NOMINATE scores, and that interestingly, *Treated* districts do not have

statistically significantly large impact on NOMINATE. We examine whether the racial composition of a district or its covered status under the VRA meaningfully predicts the ideology of its representative. Black VAP share is not a significant predictor of DW-NOMINATE scores ($\hat{\beta} = -0.110$, $p = 0.172$), and the interaction between BVAP and covered state status is similarly insignificant ($\hat{\beta} = -0.136$, $p = 0.245$). This means that packing more Black voters into a district does not seem to produce a more ideologically different representative, and this null relationship holds equally in covered and non-covered states. The one notable finding is that representatives from covered states carry a conservative baseline independent of party and district composition ($\hat{\beta} = 0.099$, $p = 0.001$), but this reflects regional political culture rather than anything specific to redistricting. Taken together, the ideology analysis suggests that while covered states pack minority voters into extreme outlier districts, the primary consequence is the structural containment of minority voting power, not a measurable shift in the ideological character of the representatives those districts produce.

Variable	Estimate	Std. Error	t	p	
Intercept	-0.37664	0.01526	-24.675	<0.001	***
Black VAP Share	-0.10961	0.08008	-1.369	0.172	
Treated	0.09890	0.02407	4.109	<0.001	***
Party	0.88153	0.01569	56.187	<0.001	***
Black VAP Share $\times$ Treated	-0.13579	0.11668	-1.164	0.245	
Observations	307				
Residual Std. Error	0.149				
F -statistic	1353***				

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Dependent variable: DW-NOMINATE first dimension score.

Table 3: BVAP Share and Ideological Extremity

5 Conclusion

The empirical results from our Difference-in-Differences analyses indicate that the *Shelby County v. Holder* (2013) decision did not fundamentally alter the mechanics of district packing in formerly covered states. The statistical insignificance of our primary DiD estimator for *max BVAP* ($\beta_3 = -2.005$, $p = 0.154$) suggests that covered states largely maintained their pre-existing redistricting strategies into the 2020 cycle. Rather than triggering a new wave of racial concentration, the removal of Section 5 preclearance appears to have had no effect on the existing packing equilibrium that was already five standard deviations removed from neutral simulation baselines in 2010, itself a legacy of the DOJ’s Max-Black strategy and the no-retrogression standard which, institutionalized super-majority districts across covered jurisdictions (Lublin 1997). The null efficiency gap result ($\beta_3 = 0.904$, $p = 0.408$) further suggests that *Shelby* did not produce a detectable shift in partisan packing either, implying that the structural geometry of these maps remained largely static across the decennial plans.

The ideological analysis reinforces the seemingly little influence of the *Shelby* decision. While

representatives from covered states are systematically more conservative than peers in non-covered states (a novel result indeed) controlling for party and district composition ($p < 0.001$), the interaction between BVAP share and covered status is insignificant ($p = 0.245$). This aligns with [Gay 2007](#)’s framework of legislating without constraints: racial packing in these states functions primarily as a mechanism of containment, cordoning off minority voting blocs into super-majority districts and liberating surrounding representatives from accountability to minority interests, rather than producing a measurable radicalization of the representatives elected from those packed districts themselves. The ideological signal is driven by party and regional alignment, not by the precise degree of packing.

The more consequential effects of *Shelby* likely lie outside district maps, as the literature does explore. As [Komisarchik and White 2020](#) demonstrate, the removal of preclearance enabled a suite of procedural changes including stricter registration requirements, polling place closures, voter ID laws, all which may widen the gap between Black and White voter turnout in formerly covered jurisdictions over time. [Trautman and Smith 2025](#) further establishes that the structural configuration of districts directly dampens grassroots mobilization and psychological electoral engagement, suggesting that the harm of the existing packing equilibrium compounds over time even absent new gerrymandering. Under the issue attention cycle ([Downs 1972](#)), the mobilization campaigns that [Cantoni and Pons 2021](#) find offset early disenfranchisement efforts are unlikely to sustain their effectiveness indefinitely as restrictive laws become further entrenched. The long-run costs of *Shelby*, in other words, may be less visible in decennial map comparisons than in the gradual erosion of participatory democracy in covered jurisdictions.

Ultimately, these results underscore that while the *Shelby* decision did not drastically worsen the state of redistricting in covered jurisdictions, the baseline reality revealed by this analysis remains deeply concerning from a fairness perspective. Covered states consistently exhibit levels of racial packing that are extreme statistical outliers relative to race-neutral simulations, concentrating minority voting power in ways that dilute political influence across a broader set of seats, precisely the dynamic that [Lublin 1997](#) identifies as the tradeoff of descriptive representation. Whether or not these configurations survive legal scrutiny in the future, they represent a structural inheritance of decades of federally incentivized concentration that *Shelby* has now left without meaningful oversight. The persistence of these outliers suggests not that the decision changed the rules of the game, but that it removed the referee, leaving in place a system in which minority communities in historically repressive jurisdictions retain significantly less political leverage than a neutral redistricting process would afford them.

Author’s Note:

No Artificial Intelligence (AI) was utilized in the ideation behind this project, the analysis of our data, or writing. However, I am bad at LaTeX, and some of the formatting was done with AI tools, specifically Claude.ai.

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